

We built a scanning tunnelling microscope in a basement

Every subsystem a tunnelling microscope needs is built, working and calibrated. The amplifier resolves picoamps, the measurement chain agrees with theory to 0.13 of a standard error, and a real tip-and-sample junction responds to voltage and to height. What is left is mechanical — and we measured it rather than guessing.

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Self-directed work: no faculty advisor, no grant and no laboratory. Mechanics, controller board and firmware follow Mech Panda's open-source *red-panda-stm*; the scan head and the transimpedance preamplifier follow Dan Berard's home-built STM. First power-on 29 August 2026; last bench session 19 September 2026. 27 session logs, more than 160 raw data files, 92 photographs of our own hardware and every analysis.

0.13σ

how far the whole measurement chain lands from what Ohm's law predicted before the run
MEASURED 2026-09-16 · panel 2

4 pA

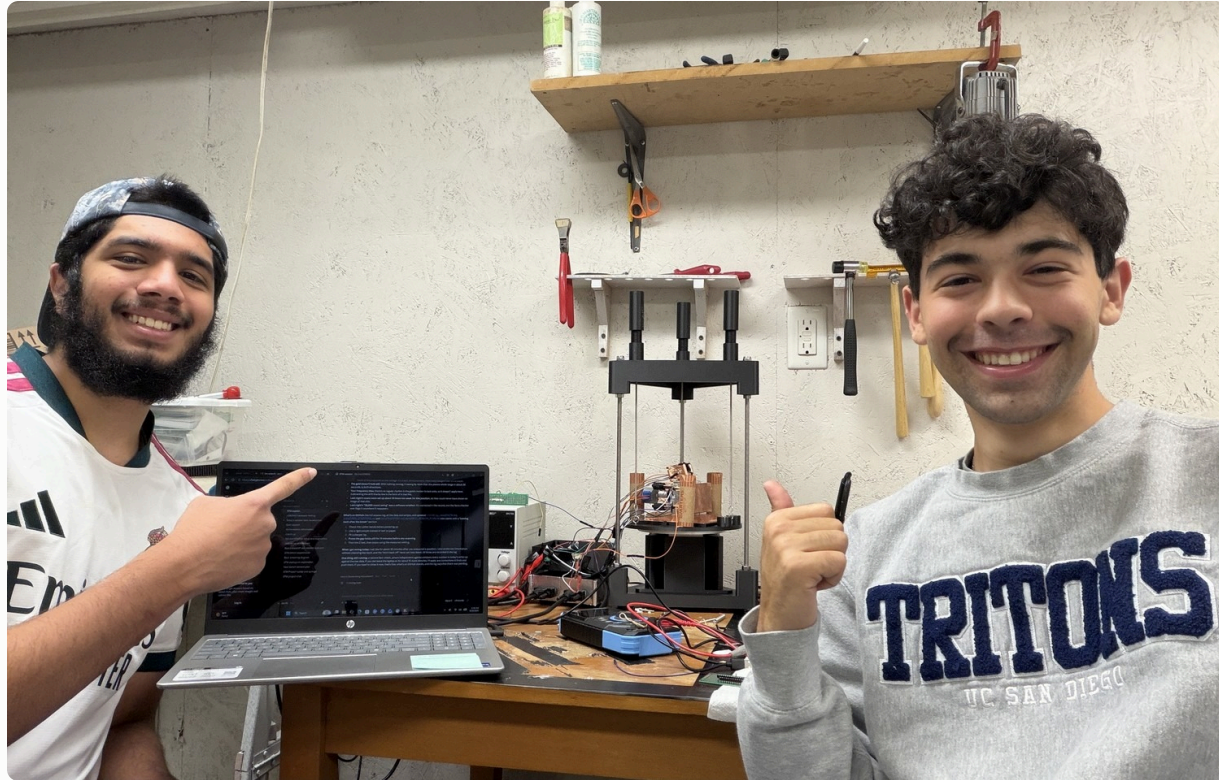
the amplifier's own input current, against the ~1 nA signal it has to find
— 250 times smaller
MEASURED 2026-09-15

16–59 MΩ

resistance of a real tip-and-sample junction across our current–voltage sweep
MEASURED 2026-09-17 · panel 3

131 → 132 pA

spread of a single reading with the room still, then with someone stamping two metres away
MEASURED 2026-09-19 · tip clear



The team, at the bench, with the instrument. Nuh Shaheer left, Jacob Katz right; the instrument is behind them, still assembled. Identification **SAID** by Jacob, 2026-09-19, who also permitted this frame to be published. Images/ours/2026-09-19.

Two first-year students. A basement. Our first soldering.

Neither of us writes code, and neither of us had much electronics experience — this project was the first real soldering either of us has done. It was built and run in the basement of a house people live in, with people moving around and no control over the air or the temperature: a wooden bench on a concrete floor, a few metres from the breaker panel. **Not a lab. Not a clean room.** (*Stated by Jacob Katz, 2026-09-19.*)

This is not a disclaimer. It is the result. A tunnelling microscope is normally a clean-room instrument — commercial ones sit on air tables in temperature-controlled labs — and the blocker we found and measured is exactly the class of problem those rooms exist to remove. Stated up front, our limitation stops being a shortfall and becomes a number for what this environment costs you.

≥ 43,000

Z counts the gap moved in 6.4 seconds with the motor stopped and nobody touching it. The blocker
MEASURED 2026-09-19 · panel 5

0 images

51 recordings and 209 reproducibility tests. No demonstrable image — and our own controls are how we know
2026-09-19 · panel 6

1 What we built

There was no kit. We designed, ordered, printed, assembled, wired and characterised a complete scanning-probe measurement chain, and brought up every stage on its own before trusting the next one.

A Teensy 4.1 drives four 16-bit converters over a 1 MHz serial bus and a 26-way ribbon — X, Y and the sample bias at ±3 V, Z at ±10 V — and their outputs move a piezoelectric disc carrying the tip. Tip current goes into an OPA627 transimpedance amplifier with a 100 MΩ feedback resistor, in a hand-folded copper box as close to the tip as it physically fits, and is read back by a 16-bit LTC2326-16. Coarse approach is a 28BYJ-48 stepper on a lever. The frame, scan head and isolation stack are 3D-printed in carbon-filled PETG; the suspended platform is a printed disc hung on three springs over an eddy-current damping stack. All control and analysis is Python on a laptop over USB.

The whole scanning module, lifted off the frame. The suspended platform with everything it carries: copper-covered base plate, the head held on by two rubber bands, the coarse-approach stepper on its lead screw, the copper-taped preamplifier box, and printed cups holding coins. IMG.4918, 2026-09-19 10:15:53, Nuh's camera. READ;

2 The chain works end to end

We took the tip and the sample out and clipped a known 100 MΩ resistor where the junction would be, then swept the bias across it. That makes the current something Ohm's law fixes **before the run**, so the instrument is checked against a number it had no part in choosing.

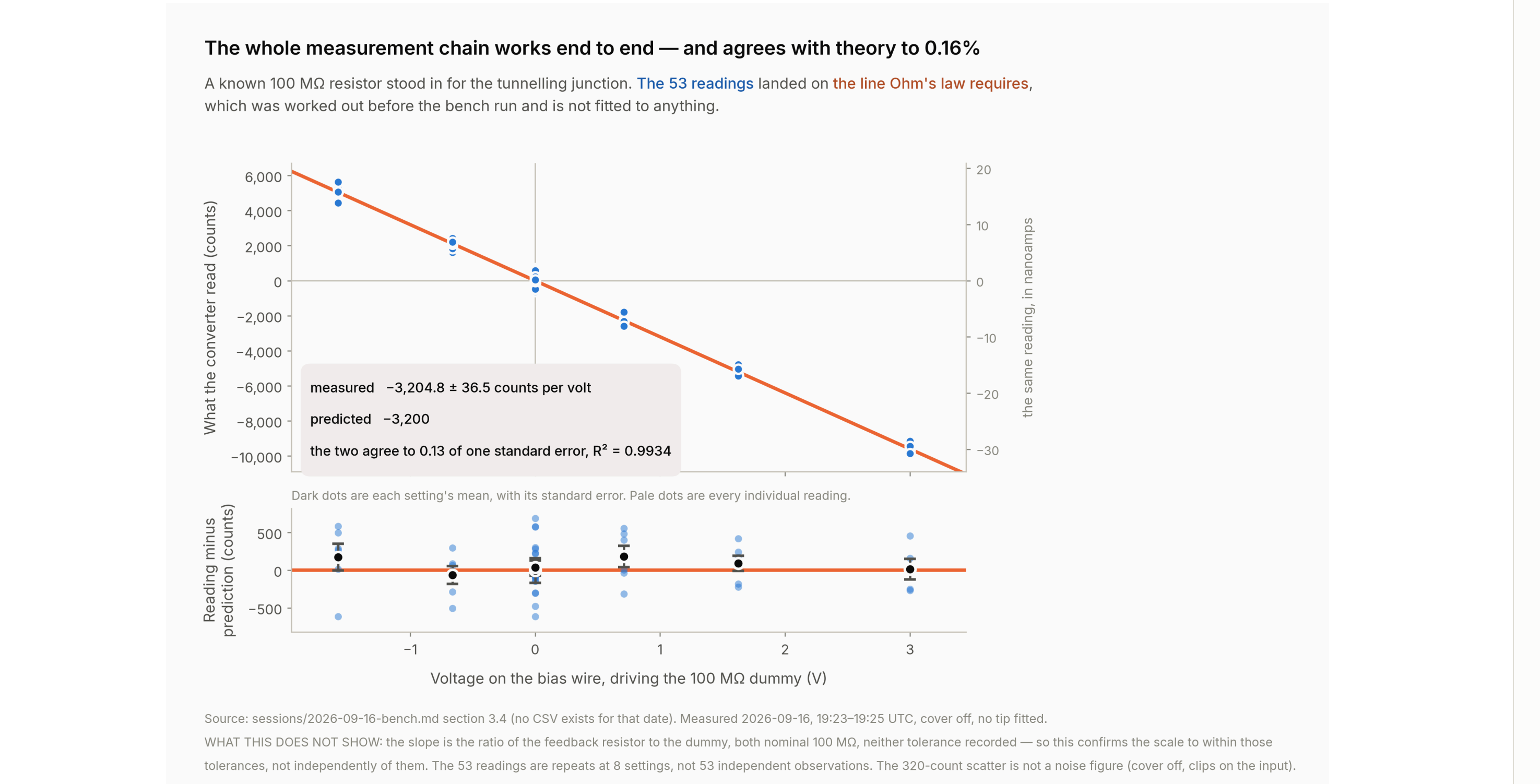


Figure 1. Measured $-3,204.8 \pm 36.5$ counts per volt against $-3,200$ predicted — agreement to 0.13 of one standard error, $R^2 = 0.9934$, 53 readings at 8 bias settings. Fifteen things had to be right at once for those points to land on that line: bias converter, buffer, wire, the known resistance, tip holder, tip lead, standoff, amplifier, cable, converter, firmware and PC tools. It also settles the sign — a positive voltage on the sample reads negative — which matters, because a current going the

5 We found what stops us, and measured it

To take an image the gap has to hold still while the tip is swept. We stopped the motor, took our hands off the instrument, and watched where the surface was.

4 What the junction is — and the one ruler that settles it

The test that separates a tunnelling gap from a contact is how fast the current changes as the tip moves. We ran it: 110 cycles in and out, at four bias voltages.

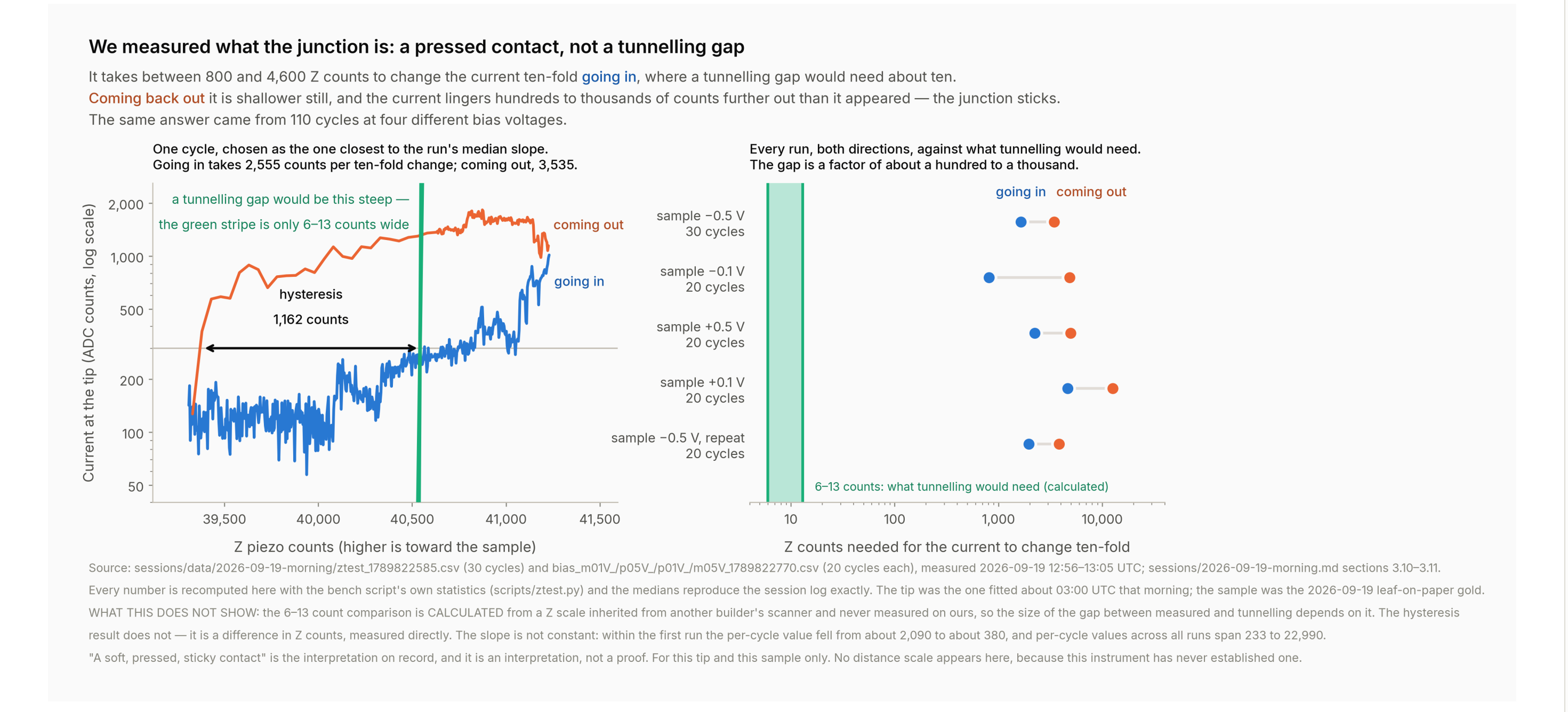


Figure 3. On the 2026-09-19 junction the current changed ten-fold per 1,650 to 1,970 Z counts going in, where a tunnelling gap on the inherited Z scale would need about 6 to 13, and every run showed 705 to 1,868 counts of in-and-out hysteresis — a mechanical signature that does not depend on any distance scale, and that did not shrink when we cut the bias fivefold. That junction was a pressed contact: measured, and not in doubt. Source: sessions/data/2026-09-19-morning/ztest and bias files, measured 2026-09-19 12:56–13:05 UTC. **What it does not show:** the 6–13 comparison is calculated from a Z scale borrowed from another builder's scanner and never measured on ours. The hysteresis does not depend on it.

And our best junction's status is still undecided — on one unmeasured distance. The junction in panel 3 is a different tip and a different sample, two nights earlier, and it changed ten-fold about seven times faster: roughly 250 Z counts per decade. Whether that was tunnelling turns on d, how far the tip stands from the line through the two ball supports, which sets the lever ratio between the motor and the tip. Below about 0.13 mm it was

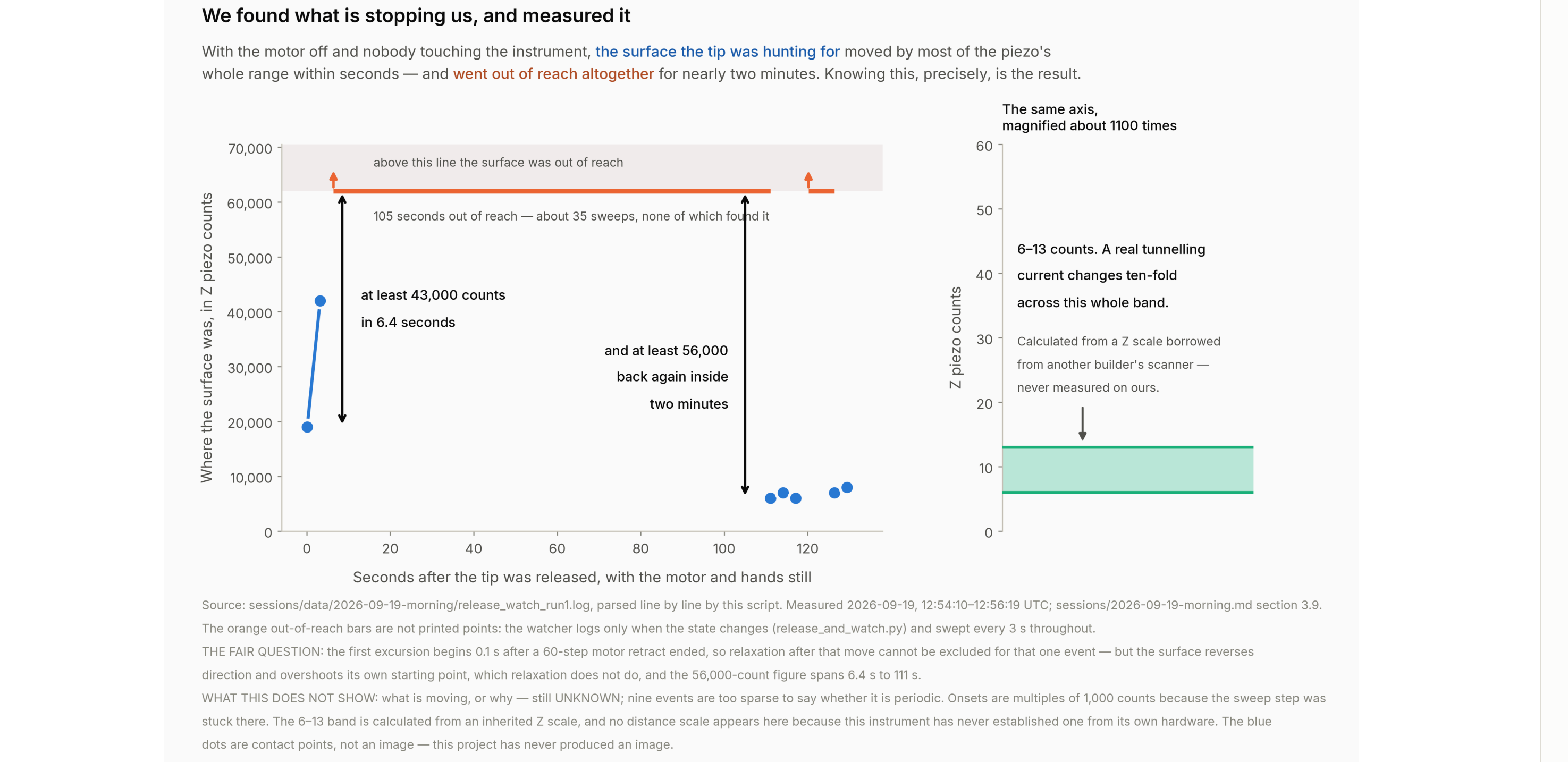


Figure 4. The surface moved by more than 43,000 Z counts within 6.4 seconds and by more than 56,000 within two minutes — most of the piezo's whole range — and went out of reach altogether for 105 seconds. Source: sessions/data/2026-09-19-morning/release_watch_run1.log, measured 2026-09-19 12:54–12:56 UTC. **The fair question:** the fast excursion begins 0.1 s after a 60-step motor retract ended, so relaxation after that move cannot be excluded for that one event — but the motion reverses and overshoots its own starting point, which relaxation does not do, and the two-minute figure is measured long after any settling transient. **Why that ends the imaging question, in one comparison:** Over the five seconds one image takes, the ran

Where it goes next

- Measure d with a straightedge — no power, a few minutes, and it settles retroactively whether our best junction was tunnelling.
- Prove the gap holds still before scanning anything: a box over the instrument, a 15-minute stillness recording, and a gate of about 200 counts over sixty seconds.
- A stiffer sample than gold leaf on paper, and a sharper tip.

3 A real junction, with the shape a barrier requires

Getting a current is easy — push the tip into the sample and there is plenty. The hard part is showing you have a barrier: two conductors with something in between that the electrons have to cross.

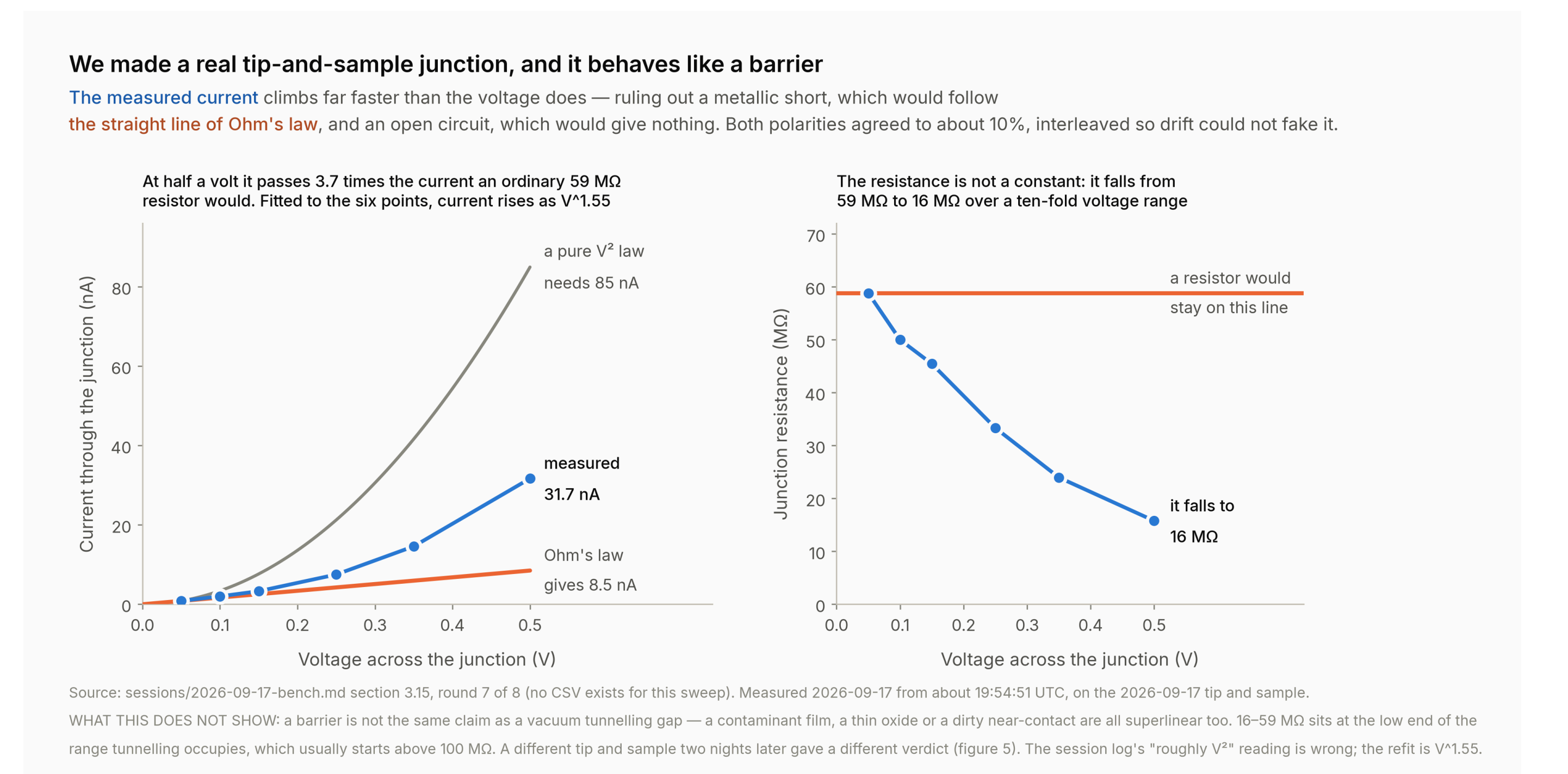


Figure 2. Current rose from 0.85 nA at 0.05 V to 31.7 nA at 0.5 V — 3.7 times faster than Ohm's law, fitting $V^{1.55}$, with the junction resistance falling from 59 to 16 MΩ. Eight rounds with the polarities interleaved so drift could not manufacture a slope, symmetric in both directions to about 10%, and confirmed by a bias-flip test on two separate nights — noise does not reverse with the voltage. That rules out a metallic short, which would be ohmic, and an open circuit, which would give nothing. Source: sessions/2026-09-17-bench.md §3.15, measured 2026-09-17. **What it does not show:** a barrier is not the same claim as a vacuum tunnelling gap — a contaminant film, a thin oxide or a dirty near-contact are all superlinear too — and 16–59 MΩ sits at the low end of the range tunnelling occupies. The session log's "roughly V^2 " reading is wrong; the refit

6 How we tested ourselves — and the one open question

Every scan was immediately re-run with the sideways sweep switched off. An X-held control has identical timing and identical loop behaviour, but the tip never travels across the surface — so it cannot contain surface structure at all, by construction. Anything appearing in both a scan and its control is the instrument talking to itself.

Like for like over full images, our feedback scans repeat at +0.04 and their controls at +0.07: indistinguishable, and neither differs from zero. That is the test that settles the imaging question, and running it was the right call.

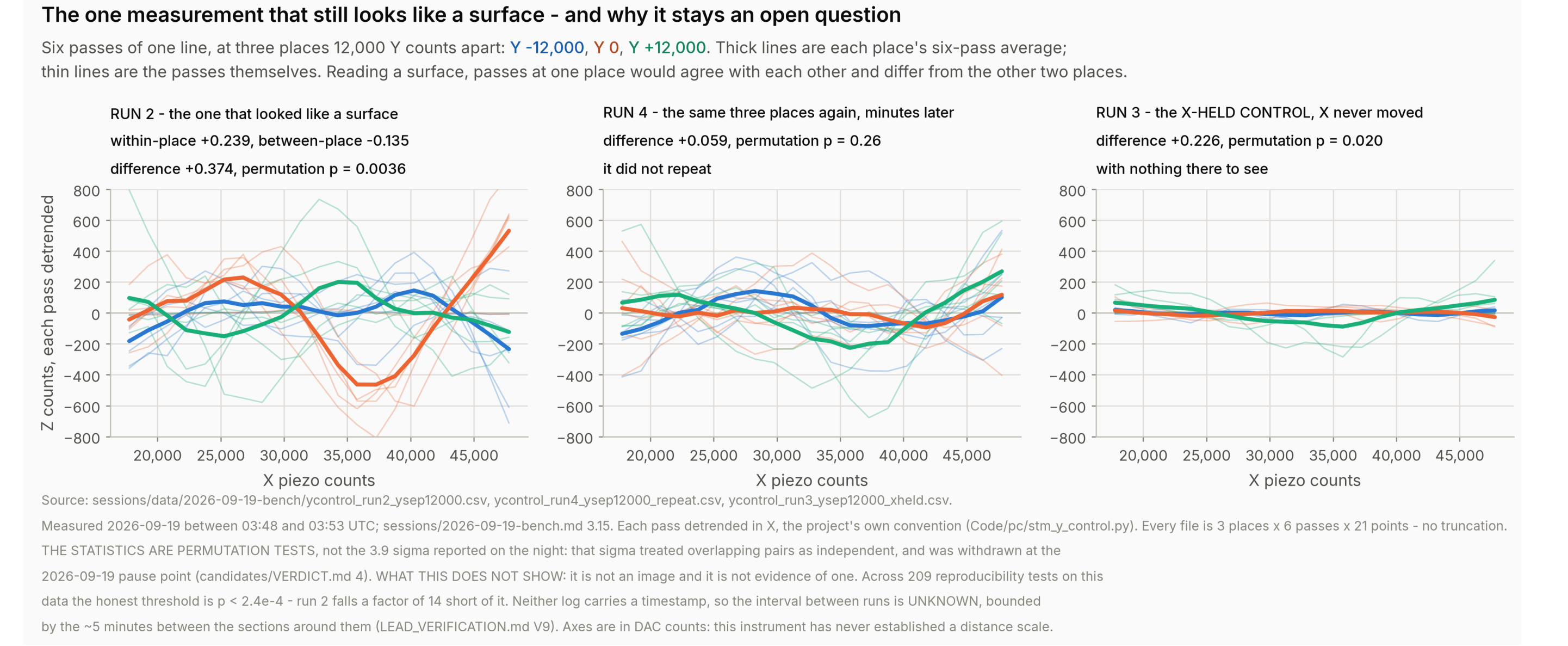


Figure 5. One measurement resists that. Holding the tip at three places 12,000 Y counts apart, passes at the same place agreed +0.239 while passes at different places disagreed -0.135 — a difference of +0.374, permutation $p = 0.0036$. The repeat, minutes later, gave +0.056 ($p = 0.26$), and an X-held control, which cannot contain a surface, scored $p = 0.020$ on the same statistic — that is this instrument's own false-positive rate, and knowing it is worth more than any single positive would have been. Across 209 reproducibility tests the honest threshold is $p < 2.4 \times 10^{-4}$, and it was fixed before the candidates were judged. Source: sessions/data/2026-09-19-bench/control_run2_run3_and_run4_measured_2026-09-19.csv, measured 2026-09-19 between 03:48 and 03:53 UTC; sessions/2026-09-19-bench.md 3.15.

We owe these people, and we checked our own work

Mech Panda (*red-panda-stm*: mechanics, controller, firmware) and Dan Berard (scan head, transimpedance preamplifier) published their instruments for free. Without them there would be nothing to bring up.

At this pause point we re-derived our own published numbers from the raw files and found six of them wrong, including two statistics we had used to dismiss our own results, and a reproducible feature that turned out to be our feedback loop

• **No number taken off a photograph.** Every caption is marked **SAID** or **READ**.

• **Three software fixes that cost nothing** and remove the artefact behind our most convincing false positive: drop the first three pixels of each pass, sweep X back and forth between passes, and refuse to compare two images of different length.

recovering from a jump rather than the sample. **Every conclusion survived; two of the arguments for them did not.** All of it is in the repository, beside the data.